



Purple Team Outline

3 Months

Month 1: Network Security & Protocol Analysis

Week 01: Network Security Lab Foundations

Objective: To initialize a professional hybrid laboratory environment and master the mechanics of low-level network communication.

- **Task 01: Virtualization & Network Topology Setup**
 - **Description:** Build the foundation of the Purple Team lab by configuring isolated attack and defense segments.
 - **Technical Requirements:**
 - Initialize Virtualization Software (VirtualBox/VMware) with dedicated Attacker (Kali) and Victim (Windows/Linux) machines.
 - Configure a strictly "Internal Network" to prevent accidental leakage while allowing inter-VM communication.
 - Verify connectivity using static IP assignments to ensure a stable testing environment.
 - **Learning Goal:** Master the creation of a safe, controlled sandbox necessary for executing high-risk security simulations.
- **Task 02: Packet Analysis & Protocol Deep-Dive**
 - **Description:** Use industry-standard tools to visualize how data moves across the wire.
 - **Technical Requirements:**
 - Perform live Packet Analysis via **Wireshark** to observe background network chatter.
 - Analyze and document the **TCP Three-Way Handshake** (SYN, SYN-ACK, ACK) in a live capture.
 - Identify common protocol headers (Ethernet, IP, TCP/UDP) and their forensic significance.
 - **Learning Goal:** Understand the fundamental "language" of the network to later identify when those protocols are being abused.

Task submission policy:

- Tasks will be assigned week-wise.
- Internees will have 1 week to complete and submit each task before the next one is assigned.
- If the previous task is not submitted, they will receive a warning.
- On the 3rd time of non-submission, they will be removed from the group.

Task submission process:

- Complete the assigned task.
- Post it on LinkedIn (as per given template).
- Must Tag Cyberster and the instructor/Mentor.
- On the 3rd time of non-submission, they will be removed from the group.
- Then share the LinkedIn post link in the respective group for review.

The task submission template should be this on LinkedIn:

- Task name
- Task by @cyberster
- Instructor name
- A little description
- Post